



Figure 1: Blockchain with smart contract

ineffective due to political/commercial reasons. Blockchain and IoT technologies together provide a solution which can do away with the intermediaries and autonomously track, control and manage resources in a trusted way.

Blockchain

The blockchain is a peer-to-peer network. It is based on a shared ledger (the blockchain database) and a protocol to keep the shared ledger updated. It is the technology underlying Bitcoin, the famous crypto-currency. Bitcoin was created by an unknown person nicknamed Satoshi Nakamoto. For people to accept and adopt Bitcoin as a crypto-currency, they had to be able to trust it. But, there is no government backing this crypto-currency. So, the trust has to be provided by the Bitcoin network itself.

The technology that provides such trust for Bitcoin is the blockchain. It is a shared ledger which is available on each full node of the network. In a Blockchain, each block is analogous to a ledger page and the blockchain is analogous to a ledger book. When a Bitcoin transaction happens on the network, the miner nodes verify the validity of the transaction, arrive at a consensus and update their copy of the ledger. The transactions verified and entered in the blockchain are immutable. Since blockchain is a peer-to-peer network, it does not have a central server. It has only peer level client applications, which join the blockchain network as nodes.

A blockchain provides some key benefits, which can be leveraged to provide solutions for use cases such as the ones discussed earlier in this article. Since it is a shared ledger on a peer-to-peer network, it does not have any central server which can be shut down unilaterally by any one party or authority. The records, transactions and smart contracts updated in the blockchain are immutable. They cannot be modified. The shared ledger is updated based on a consensus protocol and hence it prevents users or those transacting with Bitcoins from making fraudulent entries in the blockchain. This provides the trust needed when parties unknown to each other have to transact without an intermediary. Here, the blockchain network provides the trust.

There are a few open source platforms such as Ethereum, Hyperledger and Stellar, using which we can create a blockchain network as a public network, whereby any one

can join the network. Alternatively, it can also be a private network behind the firewall of an organisation so that only those within the organisation can join it; or as a permissioned blockchain, where only those who are permitted to join the blockchain can do so.

Smart contracts

The features of Blockchain, such as decentralisation, immutability, verification, consensus and not being able to shut it down, provide a trustable network which can replace any intermediary. Soon, many people realised that the blockchain has the potential to be used for applications beyond crypto-currency such as Bitcoin. It can be used as a platform to execute or enforce contracts between unknown and hence untrusted parties in the form of smart contracts. But this needed a Turing Complete blockchain platform so that complex business logic could be written as code. The Bitcoin blockchain is not Turing Complete by design to avoid hacks. Hence, smart contracts cannot be written on the Bitcoin blockchain. This limitation gave rise to other blockchain platforms which can be used to write smart contracts, such as Ethereum, Hyperledger, etc.

Smart contracts are posted on the blockchain network in the same way we send crypto-currency. The posted contracts are added to the shared ledger (blockchain database). Smart contracts implement complex business logic as code. Each smart contract will have an address to which messages can be posted. A smart contract executes upon getting the messages posted to its address.

A client application connected to the blockchain network can receive messages from real-world assets such as IoT devices and post them to the smart contract. On getting the message, the smart contract executes and sends the result back to the client application in the form of an asynchronous event. This event can be used to control/manage the real-world assets.

In a way, smart contracts are analogous to the standing instructions given to banks to perform tasks such as transferring money to another account on specific dates, paying utility bills, etc. But, smart contracts implement more complex business logic and leverage the properties of the blockchain such as decentralisation, immutability and trust.

IoT

IoT (Internet of Things) is a network of things in the physical world. These things may be devices which have sensors within them or attached to them. In the context of IoT, a 'thing' is a server which is capable of getting sensor data from the device and sending it to the backend applications. A 'thing' server can be anything – a car, a refrigerator, a machine, a surveillance camera, a fan, a light, etc. IoT uses a local gateway server to connect the various thing servers in a building, car, etc, to the backend application. Thing servers use various communication technologies such as RFID, NFC, Wi-Fi, Bluetooth, and